

Chapter 6 Series solutions of DEs

6.1 Review of Power series

A series of the form:

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

is a Power series.

More generally:

$$\sum_{n=0}^{\infty} a_n (x-c)^n = a_0 + a_1 (x-c) + a_2 (x-c)^2 + \dots$$

is a Power series centered at c.

x is a variable and the a_n 's
are the coefficients of the power series.

For any Power series :

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + \dots$$

one of the following is true.

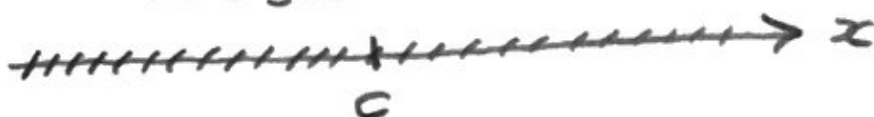
- ① The series converges absolutely for all x .
- ② The series only converges for $x=c$.
- ③ There is a number $R > 0$ called the radius of convergence such that the series converges absolutely if $|x-c| < R$ and diverges if $|x-c| > R$.

In case ① we say that $R = +\infty$.
In case ② we say that $R = 0$.

Case ①

$$R = +\infty$$

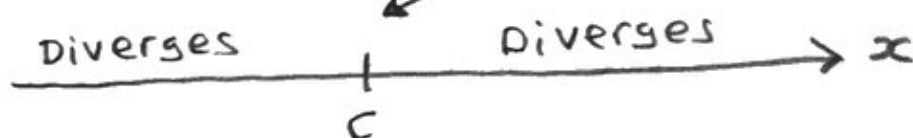
Converges for all x



Case ②

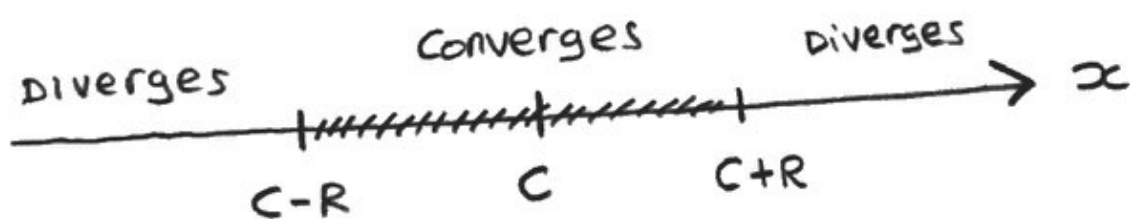
$$R = 0$$

converges only for $x = c$



Case ③

$$0 < R < +\infty$$



Convergence or divergence at $x = c - R$

and $x = c + R$ depends on the series.

The set of values of x for which the Power series converges is called the interval of convergence.

In case ③ $0 < R < +\infty$, the

interval of convergence can be any of

$$c - R < x < c + R$$

$$(c - R, c + R)$$

$$c - R \leq x < c + R$$

$$[c - R, c + R)$$

$$c - R < x \leq c + R$$

$$(c - R, c + R]$$

$$c - R \leq x \leq c + R$$

$$[c - R, c + R]$$

A power series defines a function :

$$f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n .$$

The domain of $f(x)$ is the interval of convergence of the series.

If the radius of convergence is $R > 0$ (or $R = +\infty$), then on the interval $(c-R, c+R)$ the function is differentiable. The derivative $f'(x)$ can be done termwise (as if f was a Polynomial).

$$\text{If } f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$$

$$= a_0 + a_1(x-c) + a_2(x-c)^2 + \dots$$

has a radius of convergence $R > 0$,

$$\text{then: } f'(x) = \sum_{n=1}^{\infty} n a_n (x-c)^{n-1}$$

$$= a_1 + 2a_2(x-c) + 3a_3(x-c)^2 + \dots$$

If $f(x)$ has a Power series representation centered at c , then:

$$f(x) = \sum_{n=0}^{\infty} \left[\frac{f^{(n)}(c)}{n!} \right] (x-c)^n.$$



 a_n

This is called the Taylor series of f centered at c .

If $c=0$, the series is the

Maclaurin series of f .

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \underbrace{f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots}_{\text{Maclaurin series of } f}$$

Some examples of Maclaurin series
with their interval of convergence:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad -\infty < x < \infty$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots, \quad -\infty < x < \infty$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \quad -\infty < x < \infty$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots, \quad -1 < x \leq 1$$

Definition

A function $f(x)$ is analytic at c if it can be represented as a Power series centered at c with a radius of convergence $R > 0$ (or $R = +\infty$).

In other words, $f(x)$ is analytic at c if its Taylor series centered at c has a radius of convergence $R > 0$ (or $R = +\infty$).

ex: ① $f(x) = e^x$ is analytic at 0

since its Maclaurin series ($c=0$)

Converges for all x ($R = +\infty$).

ex: ② $f(x) = \sqrt{x}$ is not analytic at $c=0$.

since $f'(x) = \frac{1}{2\sqrt{x}}$, then $f'(0)$

doesn't exist. So we can't express $f(x)$ as a Maclaurin series.

For a quotient of Polynomials

$$f(x) = \frac{P(x)}{Q(x)}$$

$f(x)$ is analytic at c

if and only if $Q(c) \neq 0$

ex: ③ $f(x) = \frac{x^2+1}{x^2-4}$ is analytic

at every x except $x = \pm 2$.

Arithmetic of Power series

Power series can be combined with addition, subtraction, and multiplication as if they were Polynomials.

$$\sum_{n=0}^{\infty} a_n x^n \pm \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (a_n \pm b_n) x^n$$

$$\left(\sum_{n=0}^{\infty} a_n x^n \right) \left(\sum_{n=0}^{\infty} b_n x^n \right)$$

$$= (a_0 + a_1 x + a_2 x^2 + \dots) (b_0 + b_1 x + b_2 x^2 + \dots)$$

$$= a_0 b_0 + (a_0 b_1 + a_1 b_0) x + (a_0 b_2 + a_1 b_1 + a_2 b_0) x^2 + \dots$$

$$= \sum_{n=0}^{\infty} c_n x^n \quad \text{with} \quad c_n = \sum_{k=0}^n a_k b_{n-k}.$$

ex: ④ Find the first three terms of the Maclaurin series of

$$f(x) = e^x \sqrt{1+x}$$

given that :

$$\left\{ \begin{array}{l} e^x = 1 + x + \frac{x^2}{2!} + \dots \\ \sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \dots \end{array} \right.$$

$$e^x \sqrt{1+x} = \left(1 + x + \frac{x^2}{2} + \dots\right) \left(1 + \frac{x}{2} - \frac{x^2}{8} + \dots\right)$$

$$= 1 + \left(x + \frac{x}{2}\right) + \left(-\frac{x^2}{8} + \frac{x^2}{2} + \frac{x^2}{2}\right) + \dots$$

$$= \boxed{1 + \frac{3x}{2} + \frac{7x^2}{8} + \dots}$$

Shifting the Summation Index

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

and

$$\sum_{k=2}^{\infty} a_{k-2} x^{k-2} = a_0 + a_1 x + a_2 x^2 + \dots$$

are the same series.

$$\sum_{n=0}^{\infty} a_n x^n = \underbrace{\sum_{k=2}^{\infty} a_{k-2} x^{k-2}}$$

Let $n = k - 2$
to recover the series
on the left.

$$k=2 \rightarrow n=0$$

ex: ⑤ Express as a single series:

$$\sum_{n=0}^{\infty} a_n x^n + \sum_{n=2}^{\infty} b_n x^n$$

Starts at 0 Starts at 2

$$a_0 + a_1 x + \sum_{n=2}^{\infty} a_n x^n + \sum_{n=2}^{\infty} b_n x^n$$

$n=0$ and $n=1$ terms of the 1st series

We can combine these two since they both start at $n=2$

$$\Rightarrow a_0 + a_1 x + \sum_{n=2}^{\infty} (a_n + b_n) x^n$$

ex: ⑥ Express as a single series

$$\sum_{n=1}^{\infty} 5n a_n x^{n-1} + \sum_{n=0}^{\infty} 3a_n x^{n+1}.$$

Step ① Express each series in powers x^k

$$\underbrace{\sum_{n=1}^{\infty} 5n a_n x^{n-1}}_{\substack{k=n-1 \\ \rightarrow n=k+1 \\ \boxed{n=1 \rightarrow k=0}}} + \underbrace{\sum_{n=0}^{\infty} 3a_n x^{n+1}}_{\substack{k=n+1 \\ \rightarrow n=k-1 \\ \boxed{n=0 \rightarrow k=1}}}$$

$$\Rightarrow \sum_{k=0}^{\infty} 5(k+1) a_{k+1} x^k + \sum_{k=1}^{\infty} 3a_{k-1} x^k$$

Step ② Combine the two series as one.

$$\sum_{k=0}^{\infty} 5(k+1)a_{k+1}x^k + \sum_{k=1}^{\infty} 3a_{k-1}x^k$$

$\downarrow$ starts at 0 $\downarrow$ starts at 1

Write explicitly the term for $k=0$ of the 1st series.

$$\underbrace{5a_1}_{k=0} + \sum_{k=1}^{\infty} 5(k+1)a_{k+1}x^k + \sum_{k=1}^{\infty} 3a_{k-1}x^k$$

1st series

$$\Rightarrow 5a_1 + \sum_{k=1}^{\infty} [5(k+1)a_{k+1} + 3a_{k-1}]x^k$$